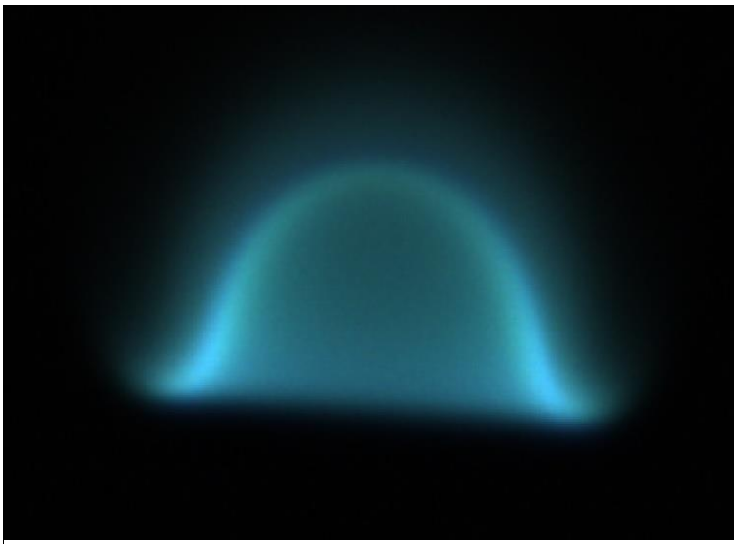
# Advanced Combustion via Microgravity Experiments (ACME)

**PURPOSE:** ACME is focused on advanced combustion technology via fundamental microgravity research. The primary goal is to improve efficiency and reduce pollutant emission in practical terrestrial combustion. A secondary objective is fire prevention, especially for spacecraft.

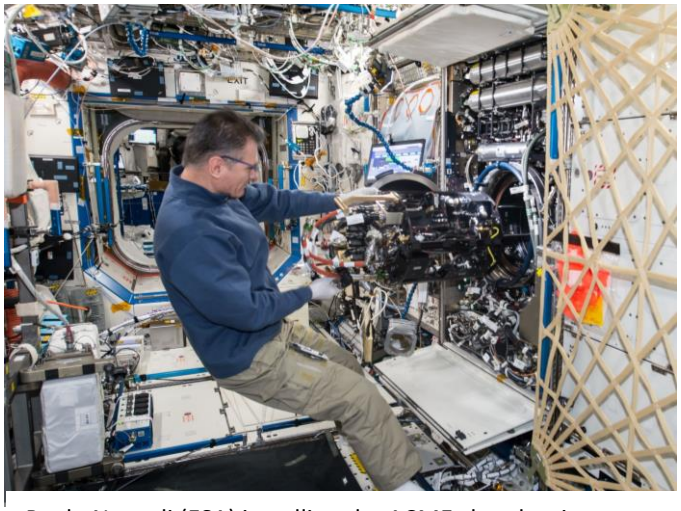
**RESEARCH:** ACME includes six independent experiments investigating laminar, gaseous, non-premixed flames. They are being conducted with a single set of modular hardware in the Combustion Integrated Rack (CIR) on the International Space Station (ISS).



Example image of a lifted coflow flame from the CLD Flame experiment of Marshall B, Long et al.

**STATUS:** ACME set-up and check-out was carried out by the ISS crew in Sept.-Oct. 2017, where science testing began in November. The first round of testing for the Coflow Laminar Diffusion Flame (CLD Flame) experiment of Marshall B. Long et al. was completed in Feb. 2018, where 130 methane flames (at a range of concentrations) were ignited. The first round of testing for the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment of Derek Dunn-Rankin et al. was completed on May 18, where 120 coflow flames were ignited. On May 20, burners and additional hardware for the next three ACME experiments are scheduled to launch. The ACME hardware should be reconfigured on May 22 by astronaut Ricky Arnold (NASA) for CLD Flame's second round, where the ethylene flame testing should begin later in the week.

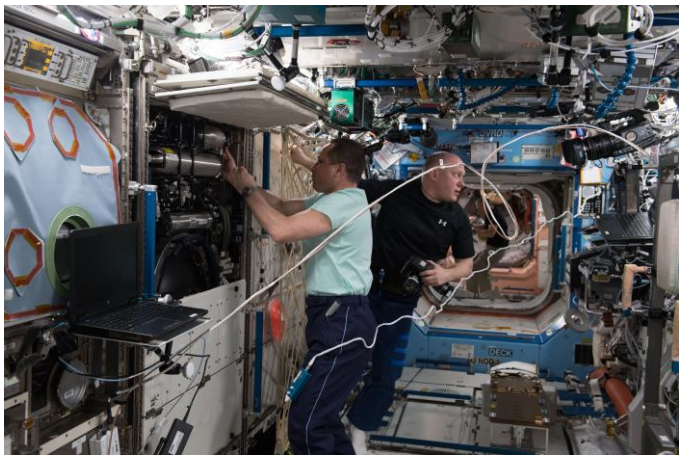
**WEB SITE:** http://spaceflightsystems.grc.nasa.gov/acme/
**UPDATES:** www.facebook.com/space.flames
**PHOTOGRAPHS:** www.flickr.com/photos/space-flames
**EDUCATION:** https://tinyurl.com/space-flames



Paolo Nespoli (ESA) installing the ACME chamber insert into the Combustion Integrated Rack (CIR)



Joe Acaba (NASA) working on the ACME chamber insert



Anton Shkaplerov (RSA) and Oleg Artemyev (RSA) exchanging a CIR gas bottle for E-FIELD Flames testing

**CONTACT:**
Dennis Stocker, ACME Project Scientist
NASA Glenn Research Center, Cleveland, OH
216-433-2166, dennis.p.stocker@nasa.gov

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